

G - Balanced Subarrays

分值: 625625625 分

Problem Statement

You are given a sequence AAA of NNN positive integers.

Among the $N(N+1)2^{\frac{N(N+1)}{2}}2N(N+1)$ non-empty (contiguous) subarrays of AAA , we call a sequence XXX a balanced sequence if and only if it satisfies the following condition:

给定一个由 NNN 个正整数组成的序列 AAA .

在 AAA 的总共 $N(N+1)2^{\frac{N(N+1)}{2}}2N(N+1)$ 个非空 (连续) 子数组中, 我们称序列 XXX 为平衡序列当且仅当它满足以下条件:

- Every integer appearing in XXX appears the same number of times in XXX .
- 所有在 XXX 中出现过的整数, 在 XXX 中的出现次数均相等。

For each $k = 1, 2, \dots, K$ $k = 1, 2, \dots, K$ $k = 1, 2, \dots, K$, find the following two values:

对于每个 $k = 1, 2, \dots, K$ $k = 1, 2, \dots, K$ $k = 1, 2, \dots, K$, 请求出以下两个值:

- The number of balanced sequences in which each integer appears exactly $B_k B_k B_k$ times.
- The number of balanced sequences in which exactly $B_k B_k B_k$ distinct integers appear.
- 满足「每个整数恰好出现 $B_k B_k B_k$ 次」的平衡序列的数量。
- 满足「恰好出现 $B_k B_k B_k$ 个不同整数」的平衡序列的数量。

Count two subarrays separately if they are taken from different positions in AAA , even if they are identical as sequences.

只要两个子数组在 AAA 中的位置不同, 即使它们作为序列完全相同, 也需要分别计数。

Solve TTT test cases per input.

每个输入包含 TTT 组测试用例, 你需要处理所有测试用例。

Constraints

- $1 \leq T \leq 2 \times 10^5$ $1 \leq T \leq 2 \times 10^5$ $1 \leq T \leq 2 \times 10^5$
- $1 \leq N \leq 2 \times 10^5$ $1 \leq N \leq 2 \times 10^5$ $1 \leq N \leq 2 \times 10^5$
- $1 \leq K \leq \min(N, 10)$ $1 \leq K \leq \min(N, 10)$ $1 \leq K \leq \min(N, 10)$
- $1 \leq A_i \leq N$ $1 \leq A_i \leq N$ $1 \leq A_i \leq N$
- $1 \leq B_k \leq N$ $1 \leq B_k \leq N$ $1 \leq B_k \leq N$
- $B_1, B_2, \dots, B_K$ $B_1, B_2, \dots, B_K$ $B_1, B_2, \dots, B_K$ are pairwise distinct.
- The sum of NNN over all test cases is at most $2 \times 10^5 \times 10^5 \times 10^5$.
- All input values are integers.
- $1 \leq T \leq 2 \times 10^5$ $1 \leq T \leq 2 \times 10^5$ $1 \leq T \leq 2 \times 10^5$
- $1 \leq N \leq 2 \times 10^5$ $1 \leq N \leq 2 \times 10^5$ $1 \leq N \leq 2 \times 10^5$
- $1 \leq K \leq \min(N, 10)$ $1 \leq K \leq \min(N, 10)$ $1 \leq K \leq \min(N, 10)$
- $1 \leq A_i \leq N$ $1 \leq A_i \leq N$ $1 \leq A_i \leq N$
- $1 \leq B_k \leq N$ $1 \leq B_k \leq N$ $1 \leq B_k \leq N$
- $B_1, B_2, \dots, B_K$ $B_1, B_2, \dots, B_K$ $B_1, B_2, \dots, B_K$ 两两不同。

- 所有测试用例的 NNN 之和不超过 $2 \times 10^5 \times 10^5 \times 10^5$ 。
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

```
TTT
case1case1case1
case2case2case2
...
caseTcaseTcaseT
```

Each case is given in the following format:

每个测试用例按照以下格式给出：

```
NNN KKK
A1A1A1 A2A2A2 ..... ANANAN
B1B1B1 B2B2B2 ..... BK BKBK
```

Output

Output the answers in the following format:

请按照以下格式输出答案：

```
$case1\mathrm{case}_1case1$
$case2\mathrm{case}_2case2$
$:\vdots:$
$caseT\mathrm{case}_TcaseT$
```

```
$case1\mathrm{case}_1case1$
$case2\mathrm{case}_2case2$
$:\vdots:$
$caseT\mathrm{case}_TcaseT$
```

For each case, letting $ck, 1c_{k,1}ck, 1$ be the number of balanced sequences in which each integer appears exactly BkB_kBk times, and $ck, 2c_{k,2}ck, 2$ be the number of balanced sequences in which exactly BkB_kBk distinct integers appear, output in the following format:

对于每个测试用例，设 $ck, 1c_{k,1}ck, 1$ 为满足「每个整数恰好出现 BkB_kBk 次」的平衡序列的数量， $ck, 2c_{k,2}ck, 2$ 为满足「恰好出现 BkB_kBk 个不同整数」的平衡序列的数量，请按照以下格式输出：

```
$c1,1c_{1,1}c1,1$ $c1,2c_{1,2}c1,2$
$c2,1c_{2,1}c2,1$ $c2,2c_{2,2}c2,2$
$:\vdots:$
$cK,1c_{K,1}cK,1$ $cK,2c_{K,2}cK,2$
```

```
$c1,1c_{1,1}c1,1$ $c1,2c_{1,2}c1,2$
$c2,1c_{2,1}c2,1$ $c2,2c_{2,2}c2,2$
$:\vdots:$
$cK,1c_{K,1}cK,1$ $cK,2c_{K,2}cK,2$
```

Sample Input 1

```
3
4 2
1 2 1 2
2 1
1 1
1
1
7 7
1 5 5 1 5 1 2
1 2 3 4 5 6 7
```

Sample Output 1

```
1 4
7 4
1 1
13 8
3 8
1 1
0 0
0 0
0 0
0 0
```

For the first test case, the pairs (l, r) such that the subarray of AAA from the l -th through the r -th element is a balanced sequence are

$(1, 1), (1, 2), (1, 4), (2, 2), (2, 3), (3, 3), (3, 4), (4, 4), (1, 1), (1, 2), (1, 4), (2, 2), (2, 3), (3, 3), (3, 4), (4, 4), (1, 1), (1, 2), (1, 4), (2, 2), (2, 3), (3, 3), (3, 4), (4, 4)$, giving eight instances in total.

Among these, the ones in which each integer appears exactly twice are $(1, 4), (1, 4), (1, 4)$, giving one instance, and the ones in which exactly two distinct integers appear are $(1, 2), (1, 4), (2, 3), (3, 4), (1, 2), (1, 4), (2, 3), (3, 4), (1, 2), (1, 4), (2, 3), (3, 4)$, giving four instances.

Thus, for $k = 1$, output 1 4.

Also, the ones in which each integer appears exactly once are

$(1, 1), (1, 2), (2, 2), (2, 3), (3, 3), (3, 4), (4, 4), (1, 1), (1, 2), (2, 2), (2, 3), (3, 3), (3, 4), (4, 4), (1, 1), (1, 2), (2, 2), (2, 3), (3, 3), (3, 4), (4, 4)$, giving seven instances, and the ones in which exactly one distinct integer appears are

$(1, 1), (2, 2), (3, 3), (4, 4), (1, 1), (2, 2), (3, 3), (4, 4), (1, 1), (2, 2), (3, 3), (4, 4)$, giving four instances.

Thus, for $k = 2$, output 7 4.

对于第一个测试用例，所有满足「 AAA 中从第 l 个元素到第 r 个元素的子数组是平衡序列」的数对 (l, r) 为

$(1, 1), (1, 2), (1, 4), (2, 2), (2, 3), (3, 3), (3, 4), (4, 4), (1, 1), (1, 2), (1, 4), (2, 2), (2, 3), (3, 3), (3, 4), (4, 4), (1, 1), (1, 2), (1, 4), (2, 2), (2, 3), (3, 3), (3, 4), (4, 4)$ ，总共有 8 个这样的子数组。

其中，满足「每个整数恰好出现 2 次」的是 $(1, 4), (1, 4), (1, 4)$ ，共 1 个；满足「恰好出现 2 个不同整数」的是

$(1, 2), (1, 4), (2, 3), (3, 4), (1, 2), (1, 4), (2, 3), (3, 4), (1, 2), (1, 4), (2, 3), (3, 4)$ ，共 4 个。

因此对于 $k = 1$ ，输出 1 4。

此外，满足「每个整数恰好出现 1 次」的是

$(1, 1), (1, 2), (2, 2), (2, 3), (3, 3), (3, 4), (4, 4), (1, 1), (1, 2), (2, 2), (2, 3), (3, 3), (3, 4), (4, 4), (1, 1), (1, 2), (2, 2), (2, 3), (3, 3), (3, 4), (4, 4)$ ，共 7 个；满足「恰好出现 1 个不同整数」的是 $(1, 1), (2, 2), (3, 3), (4, 4), (1, 1), (2, 2), (3, 3), (4, 4), (1, 1), (2, 2), (3, 3), (4, 4)$ ，共 4 个。

因此对于 $k = 2$ ，输出 7 4。